Markets

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Chapter 2: Economic Models

How Economists Study the World

- Conceptual analysis: Develop hypotheses
- Economic modeling: Simplify reality and clarify reasoning
- Empirical analysis: Use econometrics to test hypotheses
- Historical and institutional analysis
- Experimental analysis: Examine hypotheses in controlled lab or field settings

Models in Economics

- Models are simplified representation of reality.

- They make assumptions and reasoning explicit and transparent.

- They enable comparative statics.

Ceteris paribus: other things equal.

Models in Economics

"[A]ll models are wrong, but some are useful…

…[T]he question is how wrong do they have to be [be dangerous!]".

— George E.P. Box

Forms of Models

- Diagrams: intuitive, visual, easy to apply economic intuition
- Math: precise, rigorous, and can yield results beyond initial intuition
- Principles of Economics (Markets): primarily diagrammatic;
- Intermediate Micro/Macro: a balanced use between diagrams and mathematical formulas;
- Advanced Micro/Macro: primarily mathematical.

Examples of Models

(I) Production Possibility Frontier

(II) Gains from Trade

(III) Circular-Flow Diagram (self-study)

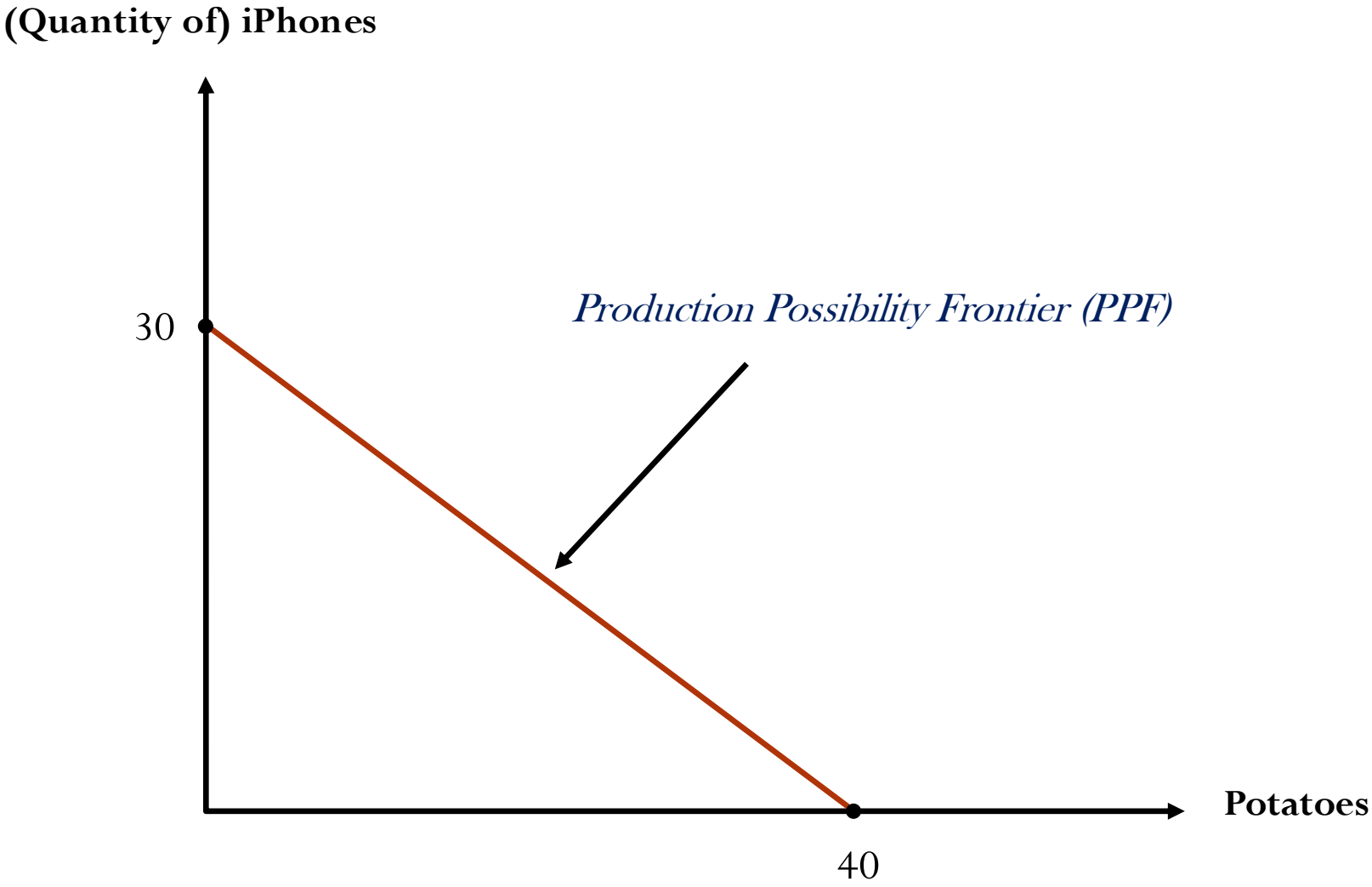
Model I: Production Possibility Frontier

- Suppose Alice has **10 hours** of working time.
- Each hour can be used to produces either iPhones and potatoes.

	iPhones	potatoes
Per hour productivity	3	4

- How can we describe Alice's production possibilities succinctly?

Alice's Production Possibilities

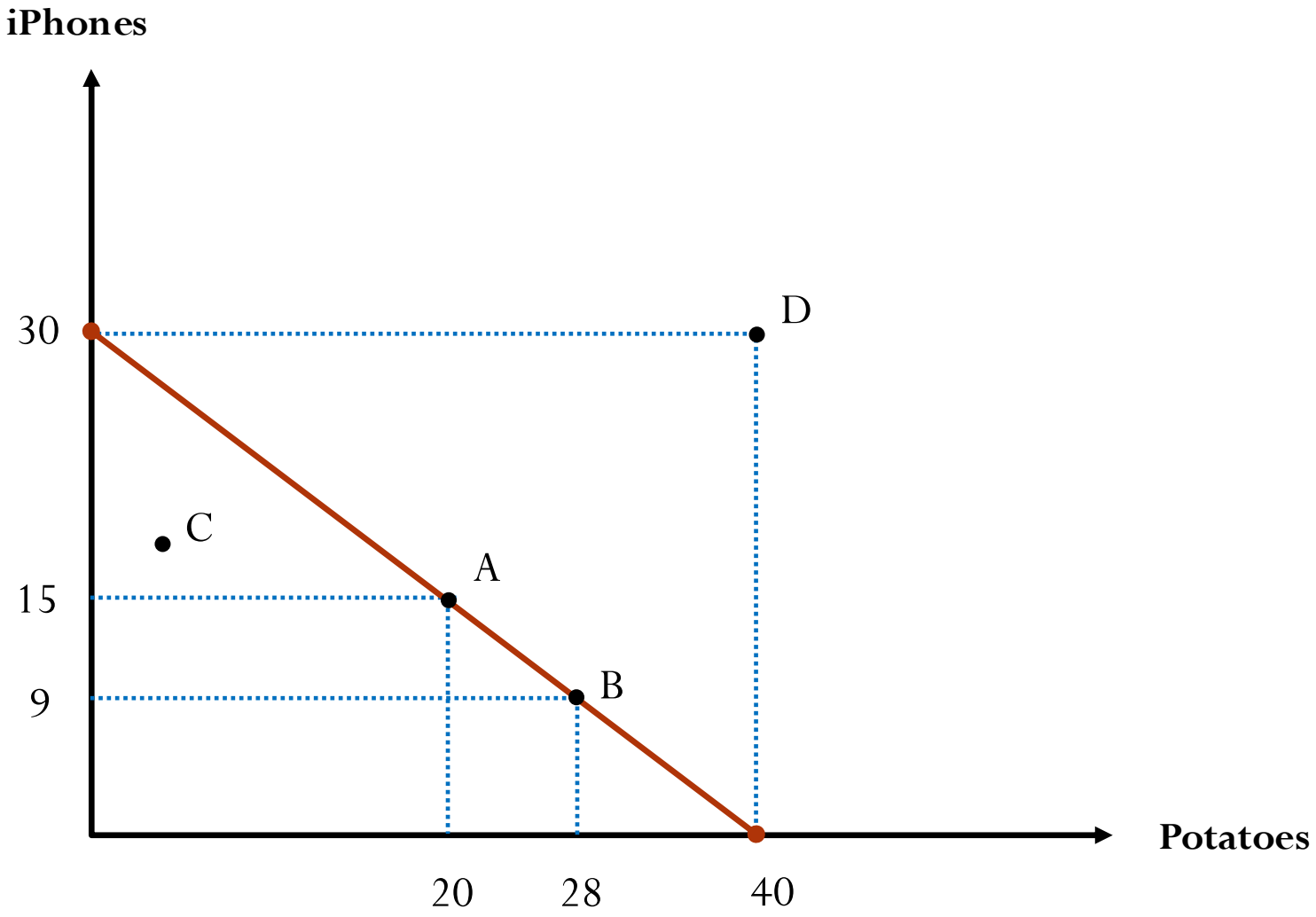


Production Possibility Frontier

Given the production technology and resources, the PPF shows the maximum combinations of outputs that can be produced.

- Could the PPF slope upward?
- What can we learn from the PPF?

Feasibility and Efficiency



Opportunity Cost

- Suppose Alice decides to change its production from point A to point B – it will produce 8 more potatoes but 6 fewer iPhones.

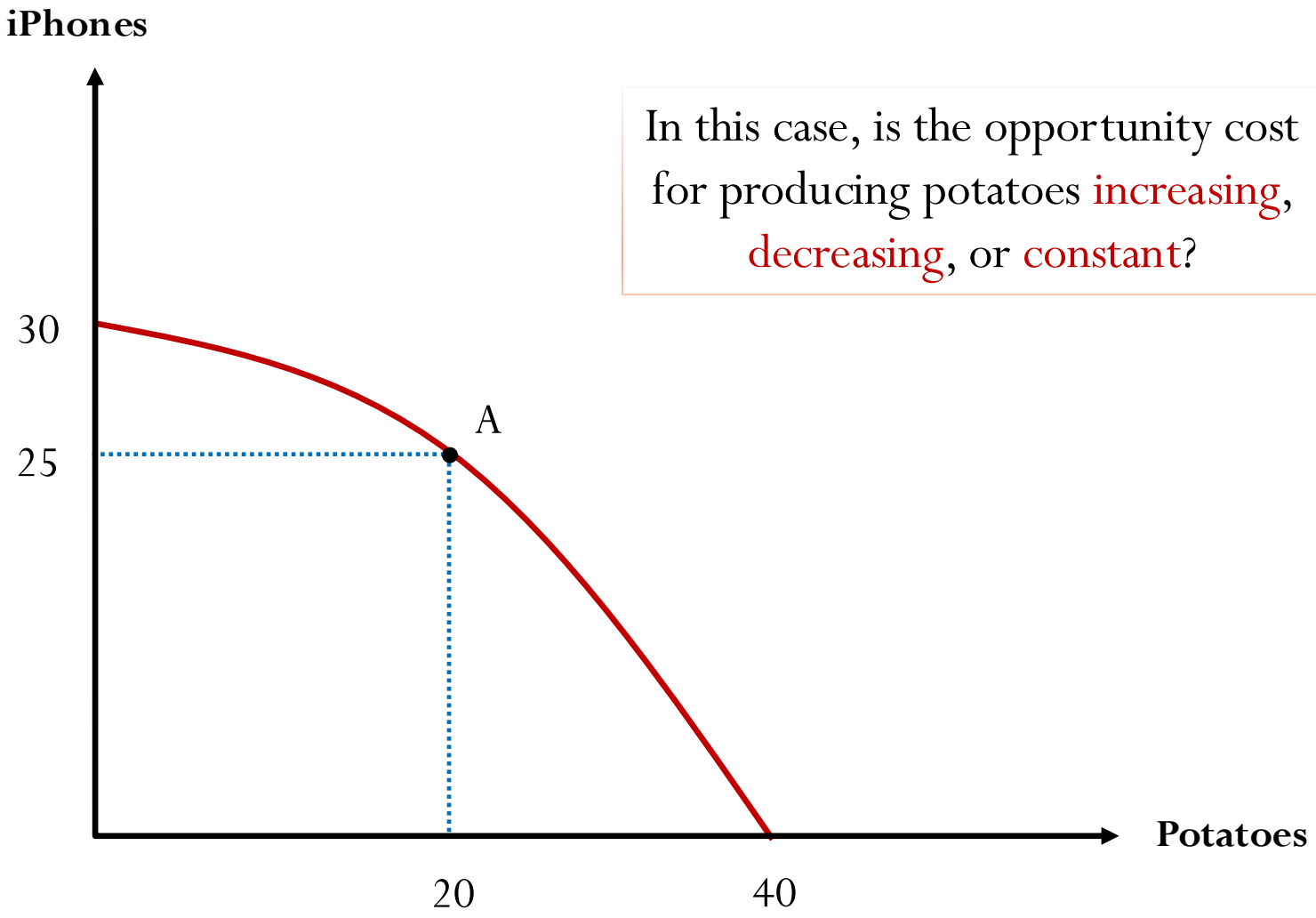
- Thus, the opportunity cost of a potatoes in terms of iPhones is

$$\frac{\text{what is given up}}{\text{what is gained}} = \frac{6}{8} = \frac{3}{4}$$

- Because the trade-off is the same for every potato, the opportunity cost is constant.

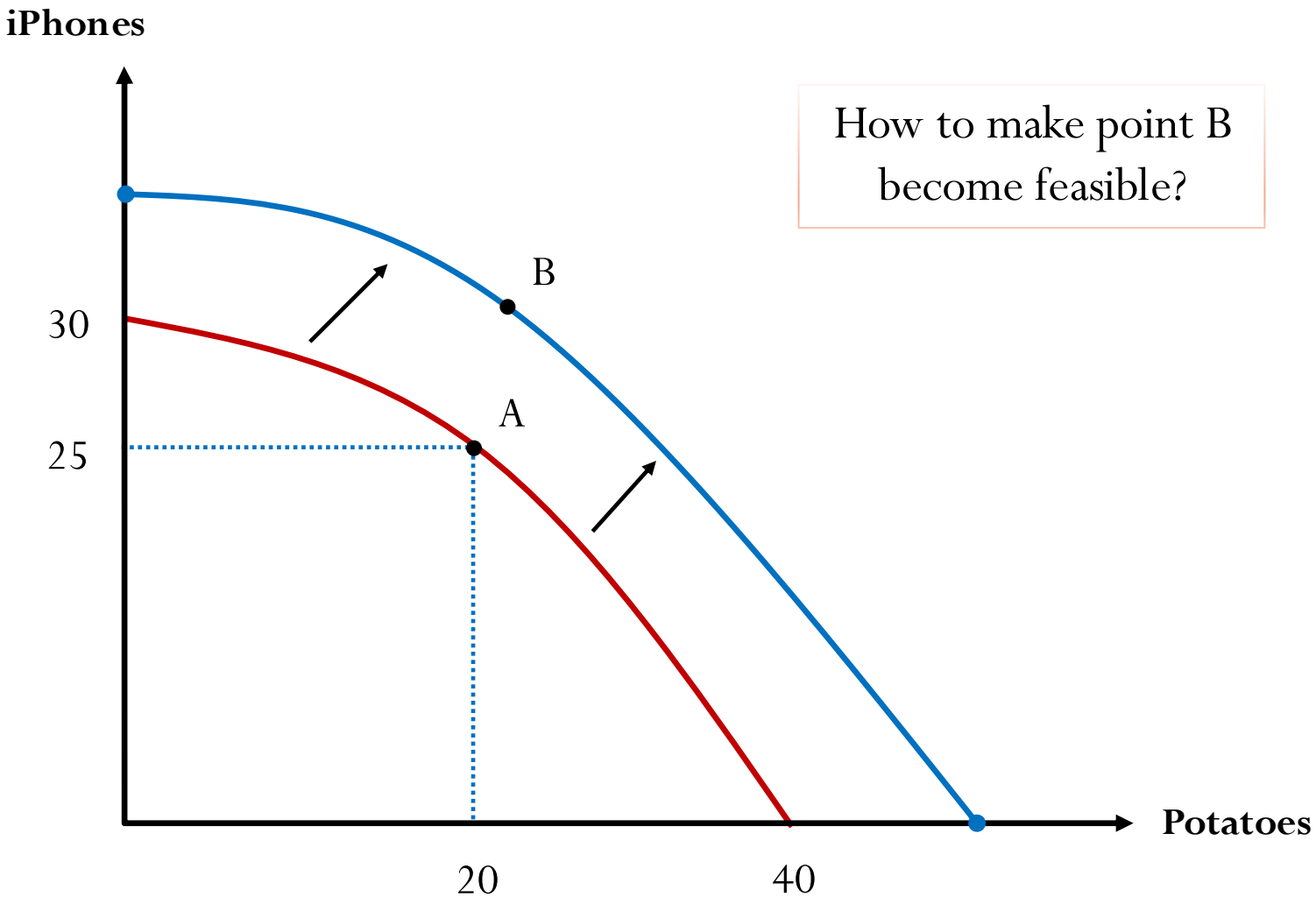
- This is represented by a linear PPF.

Opportunity Cost



In this case, is the opportunity cost for producing potatoes increasing, decreasing, or constant?

Economic Growth



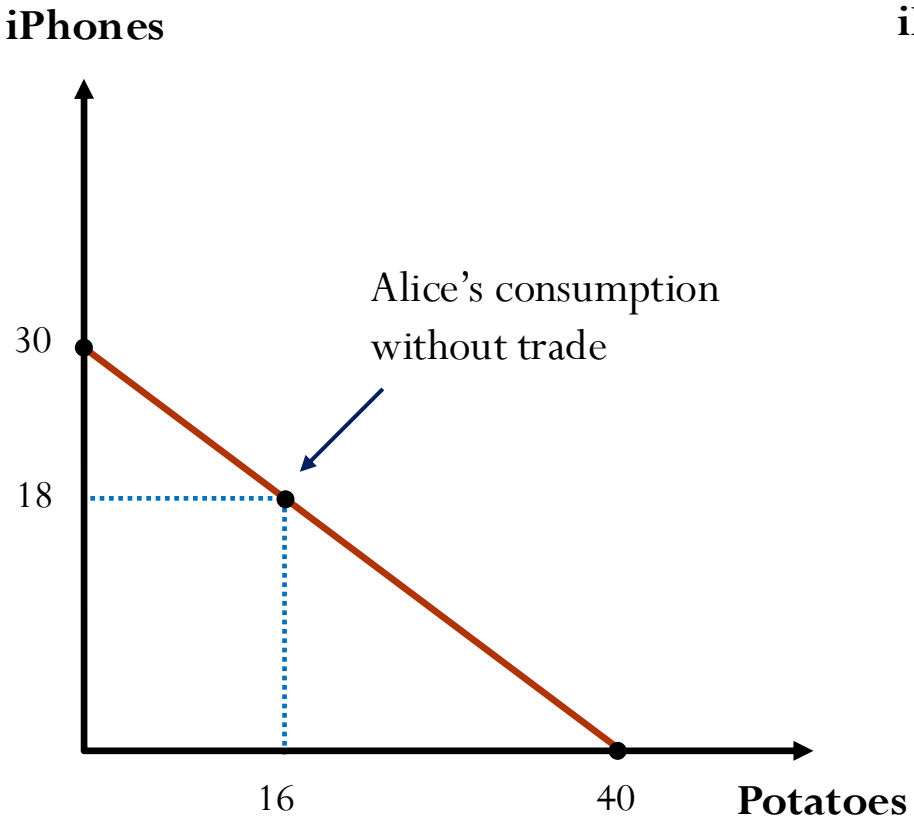
How to make point B become feasible?

Model II: Gains from Trade

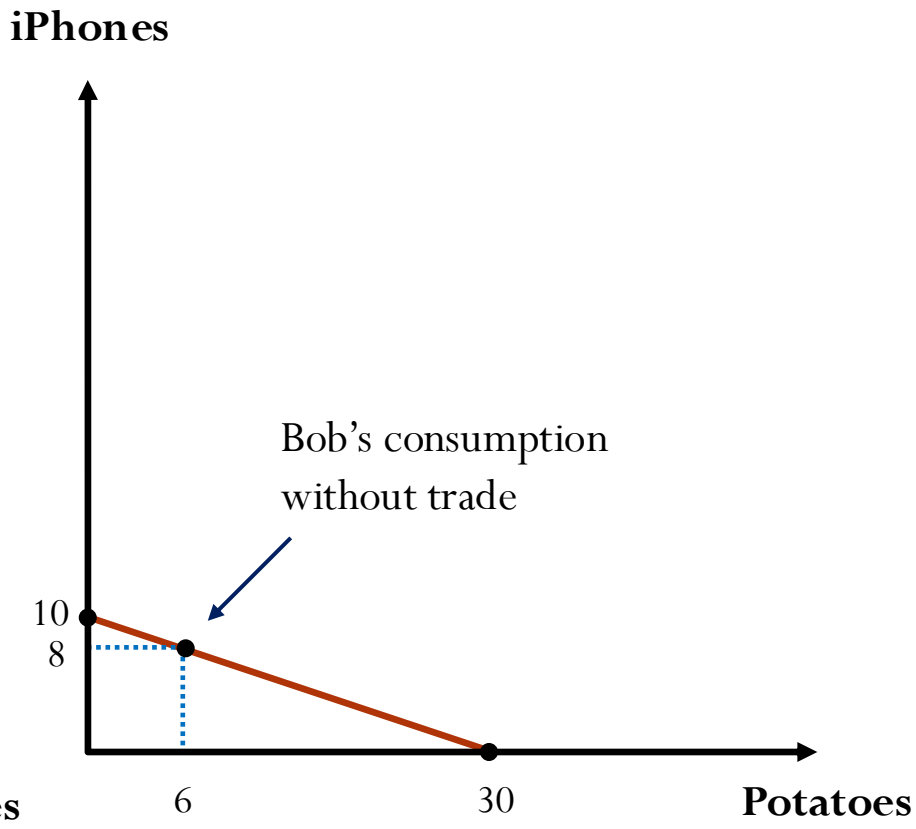
- Recall Principle 5: There are gains from trade.
- But how should we model such gains?
- A good model should help us answer three key questions:
  - If no trade is allowed, how would parties organize production and consumption on their own?
  - If trade is allowed, how would production be arranged before exchange takes place, …
  - … and how would consumption be allocated after exchange?

A Tale of Two Producers

(a) Alice's production and consumption



(b) Bob's production and consumption



Opportunity Costs

	Alice's opportunity cost	Bob's opportunity cost
1 potato	? iPhones	? iPhones
1 iPhone	? potatoes	? potatoes

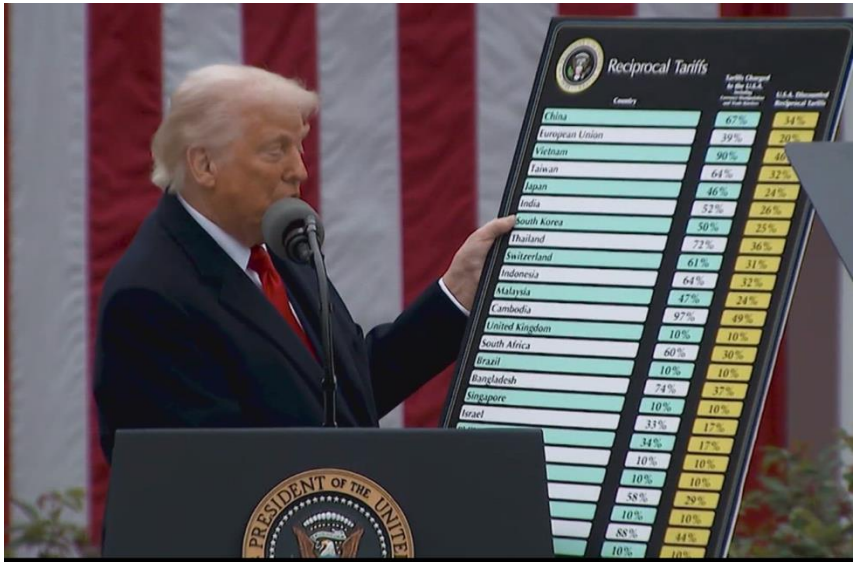
Opportunity Costs

	Alice's opportunity cost	Bob's opportunity cost
1 potato	3/4 iPhones	1/3 iPhones
1 iPhone	4/3 potatoes	3 potatoes

A Trade Deal

Imagine a trade deal is reached:

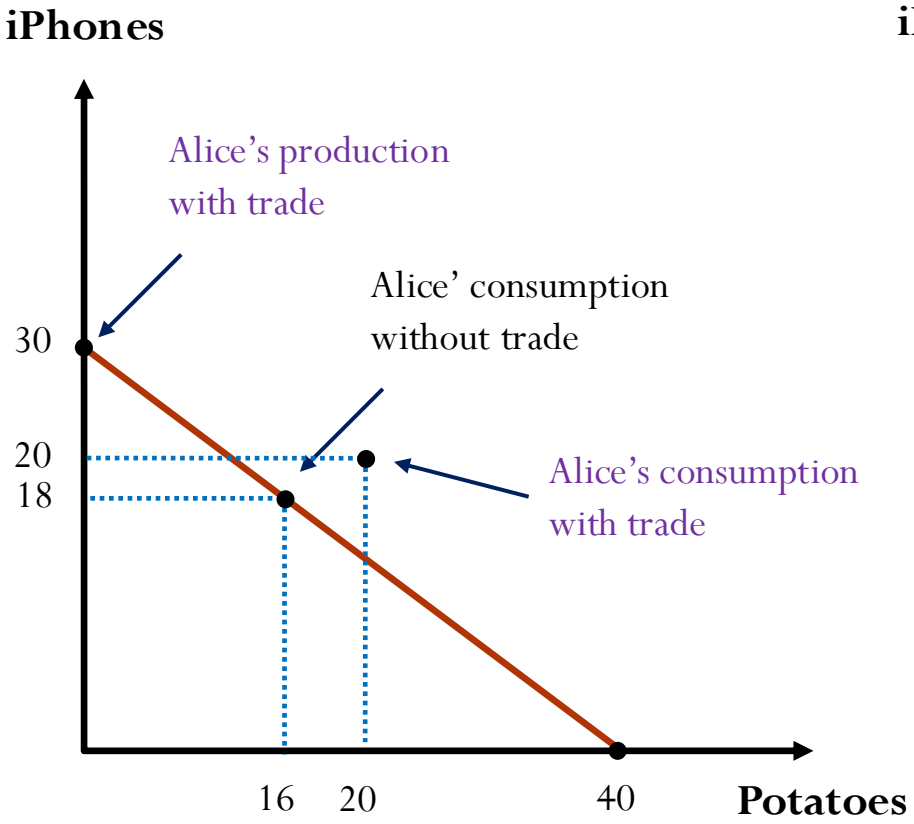
- Alice agrees to specialize in producing iPhones;
- Meanwhile, Bob will focus on producing potatoes.
- They then exchange.
- Say, Alice gives10 iPhones in return for 20 potatoes from Bob.



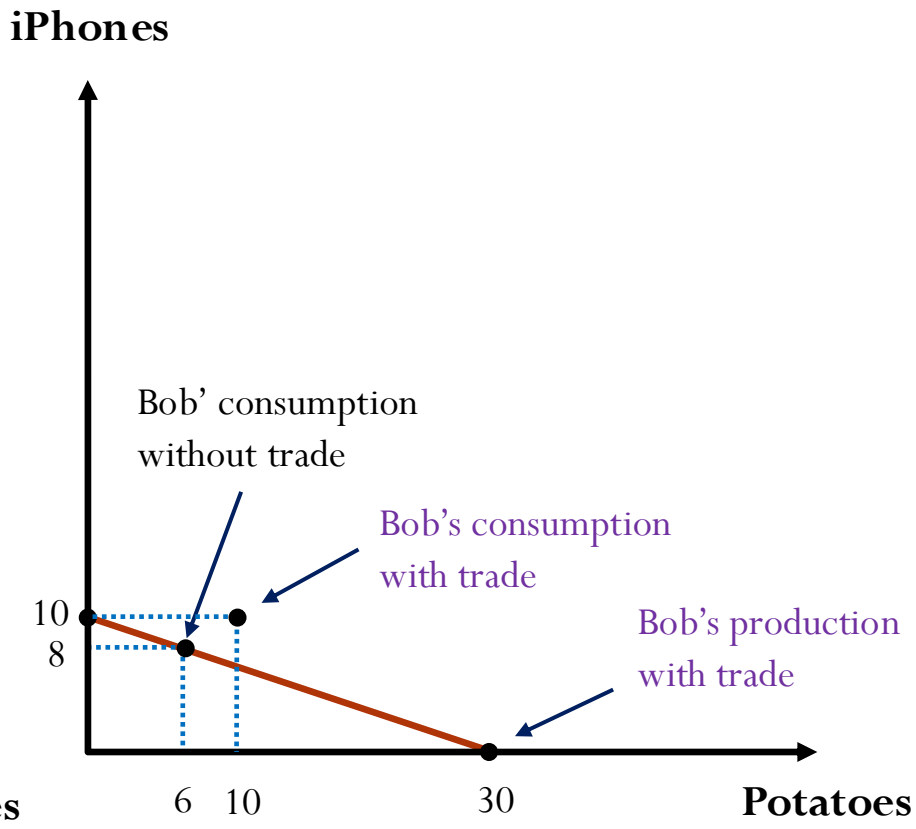
(Not this kind…)

Voila!

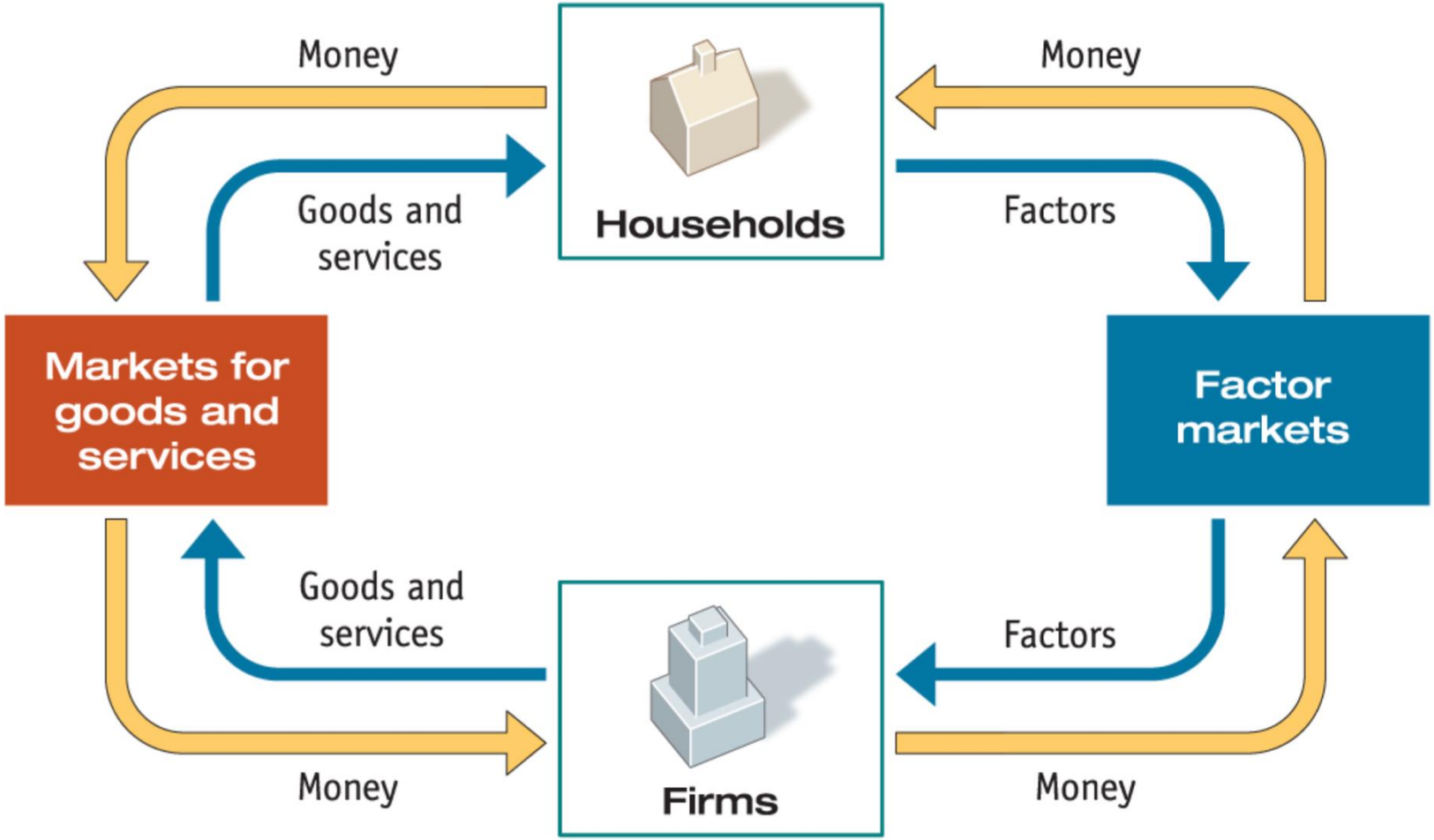
(a) Alice's production and consumption



(b) Bob's production and consumption



Model III: Circular-Flow Diagram



Using Models: Positive versus Normative

- **Positive economics**
  - Describes economic outcomes as they are;
  - Focuses on explaining and predicting how the economy actually works (or would work)
- **Normative economics**
  - Expresses value judgements about how the economy should work
  - Provides guidance and recommendations for policy or decision-making

Positive or Normative?

I. Global temperatures are projected to rise between 2°C to 5°C under high growth scenarios.

II. CO_2 emissions should be reduced, as global GDP in 2010 is expected to be 25% lower if temperatures rise by 4°C compared to no increase.

III. Raising the minimum wage would lower the employment rate of unskilled workers.

IV. Raising the minimum wage can improve overall social welfare.

When and Why Economists Disagree?

- Models help structure economic reasoning, but economists may still agree or disagree.
- Broad agreement:
  - ❑ Very few investors can predict stock prices — 100%
  - ❑ Raising the federal tax rate will increase tax revenue — 95%
  - ❑ On average, NAFTA benefits U.S. citizens — 85%
  - ❑ The 2009 stimulus reduced unemployment — 89%
  - ❑ Carbon taxes are better than emission standards — 88%
  - ❑ Healthcare benefits cannot be sustained without tax increases — 85%
  - ❑ Changes in gas prices are mostly due to market factors — 85%

When and Why Economists Disagree?

- Models help structure economic reasoning, but economists may still agree or disagree.
- More divided views:
  - The benefits of bailing out car manufacturers exceeded the costs — 40% (30% uncertain)
  - School vouchers improve student outcomes — 29% (43% uncertain)
  - The benefits of the 2009 stimulus exceeded the costs — 35% (33% uncertain)
  - Large banks would not be smaller without government support — 29% (56% uncertain)
  - CEOs are not overpaid — 29% (52% uncertain)

When and Why Economists Disagree?

- Disagreement on positive issues often stems from choosing different models or assumptions.
- Because positive claims can be tested against data, some models may ultimately prove more accurate or applicable.
- In contrast, normative statements rest on value judgements, and economists need not assign the same weights to different outcomes.
- Yet, even when making normative claims, economists typically draw on empirical evidence.

Summary

- Introduction to economic models
- Examples: PPF, gains from trade, circular-flow diagram
- Positive versus normative economics
- Appendix (please do study yourself): graphs in economics — plotting curves, calculating slope, measuring areas, types of graphs, etc
- Next lecture, Chapter 3: Supply and Demand